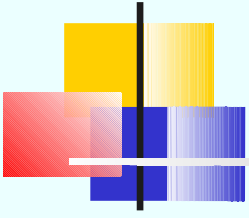


Programming with Android

Android advance

Dr. Nguyen Manh Hung

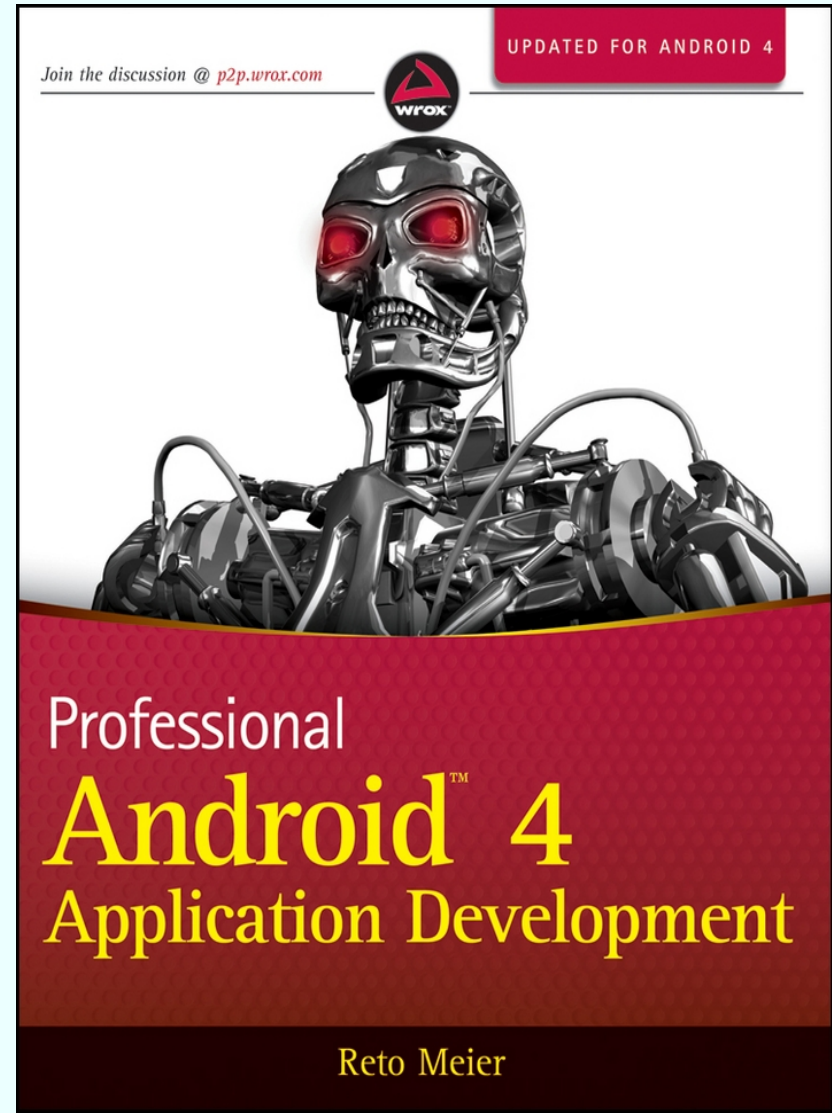
Posts & Telecommunications Institute of Technology (PTIT)

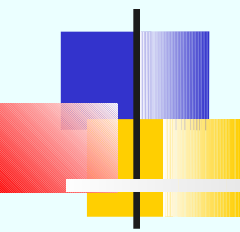


Content

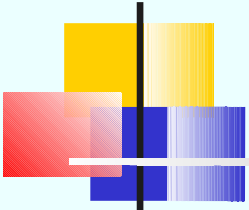
- Facebook integration
- Google maps
- Image
- Internal storage
- JSON, RSS Parser
- Localization
- Loading spinner
- Network connection

References





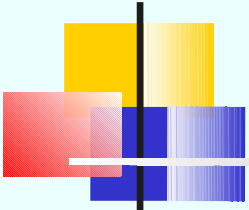
Facebook integration



Facebook integration

Two ways:

- Facebook SDK
- Intent Share



Facebook SDK (1)

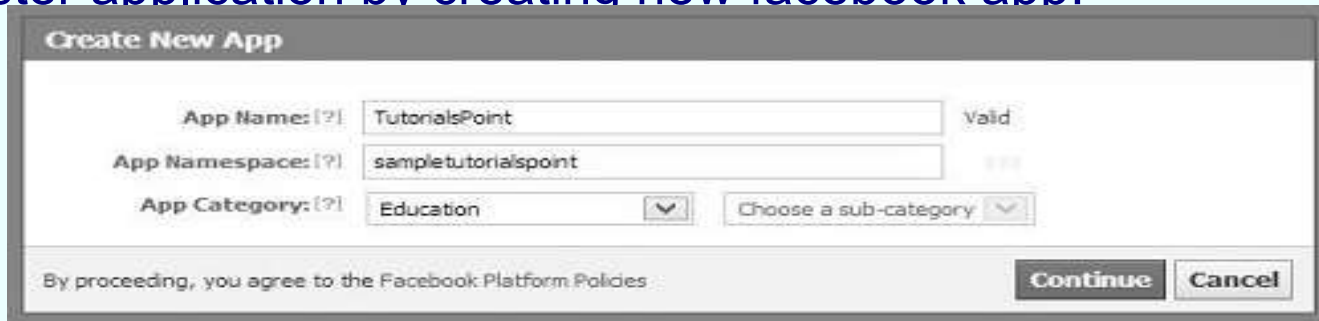
Generating application signature:

```
keytool -exportcert -alias androiddebugkey -keystore "your path" |  
openssl sha1 -binary | openssl base64
```

```
C:\Program Files\Java\jre7\bin>keytool -exportcert -alias androiddebugkey -keystore "C:\Users\      \.android\debug.keystore" |  
openssl sha1 -binary | openssl base64  
Enter keystore password: android  
gr0ix/+LES0xzIwE
```

Facebook SDK (2)

Register application by creating new facebook app:



Create New App

App Name: [?] TutorialsPoint Valid

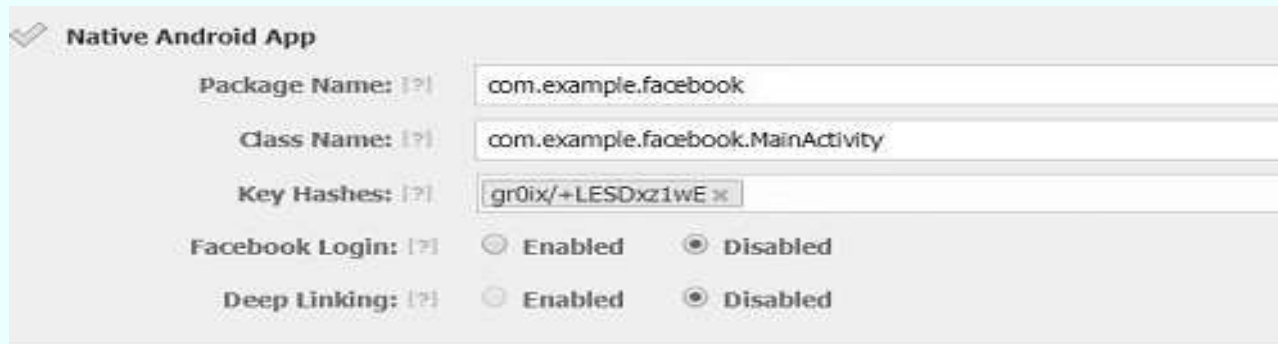
App Namespace: [?] sampletutorialspoint

App Category: [?] Education Choose a sub-category

By proceeding, you agree to the Facebook Platform Policies

Continue **Cancel**

In the code of project in Eclipse:



☒ **Native Android App**

Package Name: [?] com.example.facebook

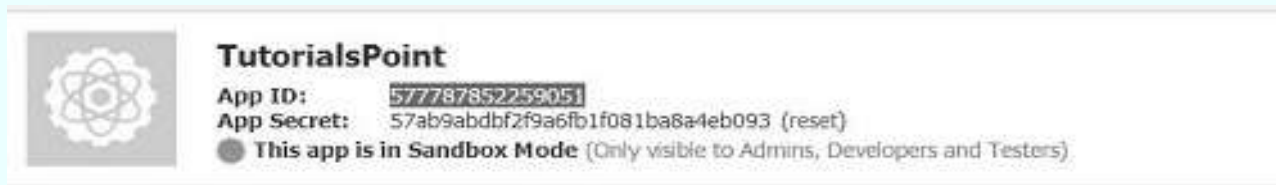
Class Name: [?] com.example.facebook.MainActivity

Key Hashes: [?] gr0ix/+LESDxz1wE

Facebook Login: [?] ☐ Enabled ☒ Disabled

Deep Linking: [?] ☐ Enabled ☒ Disabled

Receiving app id:

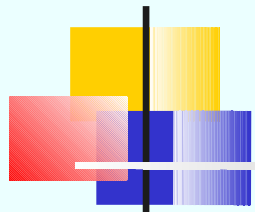


TutorialsPoint

App ID: 577787852259051

App Secret: 57ab9abdbf2f9a6fb1f081ba8a4eb093 (reset)

This app is in Sandbox Mode (Only visible to Admins, Developers and Testers)

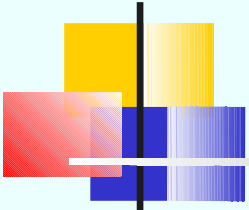


Facebook SDK (3)

Download Facebook SDK:

<https://github.com/facebook/facebook-android-sdk>

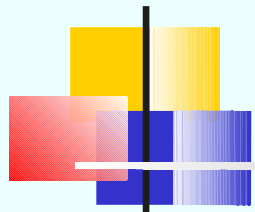
Import it into the project in Eclipse



Create facebook login

code:

```
// start Facebook Login
Session.openActiveSession(this, true, new Session.StatusCallback() {
// callback when session changes state
public void call(Session session, SessionState state, Exception
exception) {
    if (session.isOpened()) {
        // make request to;2 the /me API
        Request.executeMeRequestAsync(session, new Request.
            GraphUserCallback() {
                // callback after Graph API response with user object
                @Override
                public void onCompleted(GraphUser user,
                    Response response) {
                    if (user != null) {
                        TextView welcome = (TextView)findViewById(R.id.welcome);
                        welcome.setText("Hello " + user.getName() + "!");
                    }
                }
            }
        ));
    }
}
```



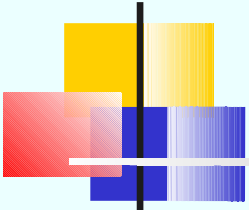
Intent Share

Create:

```
Intent shareIntent = new Intent();  
shareIntent.setAction(Intent.ACTION_SEND);
```

Using:

```
shareIntent.setType("text/plain");  
shareIntent.putExtra(Intent.EXTRA_TEXT,  
    "Hello, from tutorialspoint");  
startActivity(Intent.createChooser(shareIntent,  
    "Share your thoughts"));
```

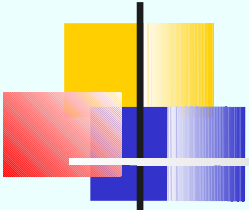


Example (1)

MainActivity.java:

```
class MainActivity extends Activity {
    private ImageView img;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        img = (ImageView) findViewById(R.id.imageView1);
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }

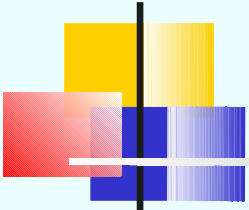
    public void open(View view) {
        Intent shareIntent = new Intent();
        shareIntent.setAction(Intent.ACTION_SEND);
        shareIntent.setType("text/plain");
        shareIntent.putExtra(Intent.EXTRA_TEXT,
            "Hello, from tutorialspoint");
        startActivity(Intent.createChooser(shareIntent,
            "Share your thoughts"));
    }
}
```



Example (2)

activity_main.xml:

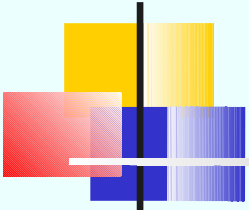
```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
<ImageView
android:id="@+id/imageView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentLeft="true"
android:layout_alignParentTop="true"
android:layout_marginLeft="98dp"
android:layout_marginTop="139dp"
android:onClick="open"
android:src="@drawable/tp" />
<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="48dp"
android:text="@string/tap"
android:textAppearance="?android:attr/textAppearanceLarge" />
</RelativeLayout>
```



Example (3)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <string name="app_name">IntentShare</string>
  <string name="action_settings">Settings</string>
  <string name="hello_world">Hello world!</string>
  <string name="tap">Tap the button to share something</string>
</resources>
```

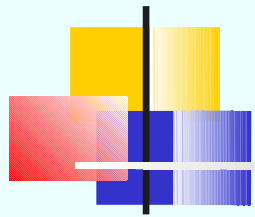


Example (4)

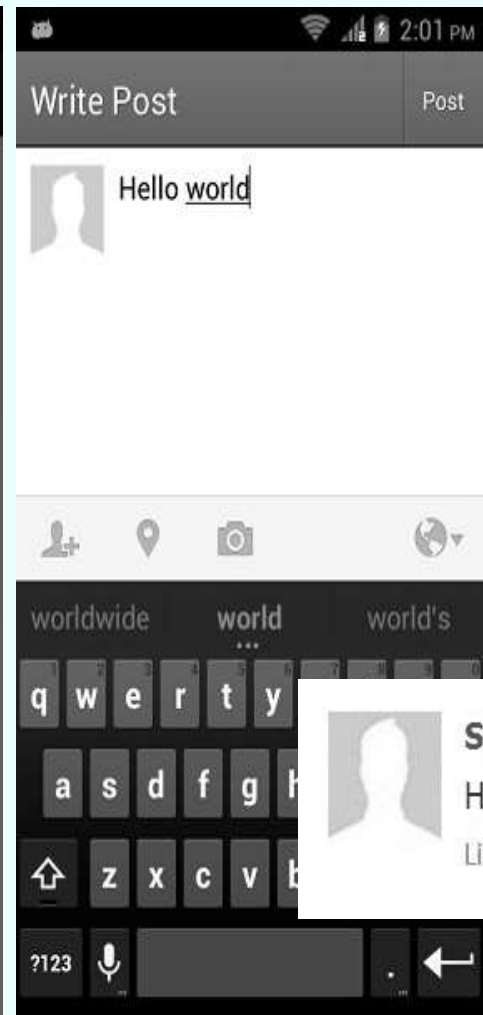
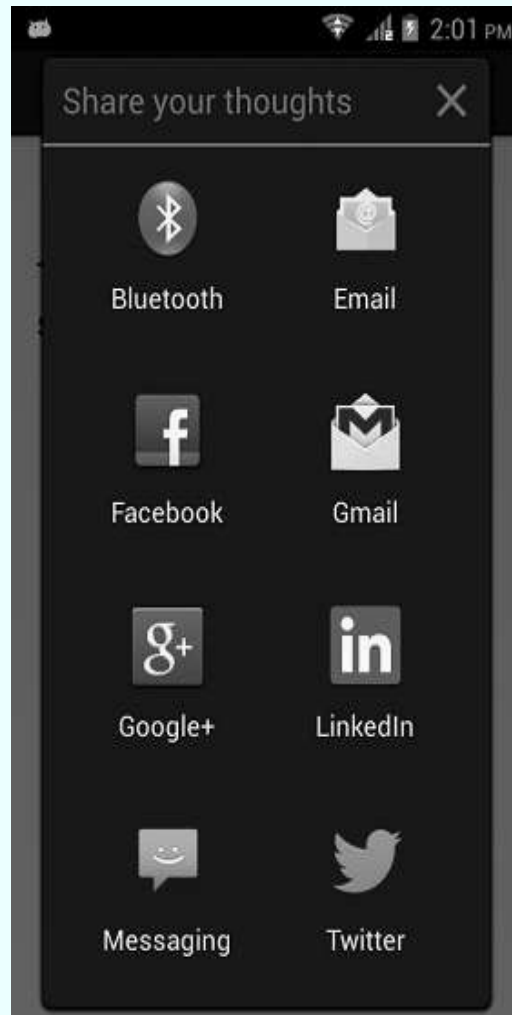
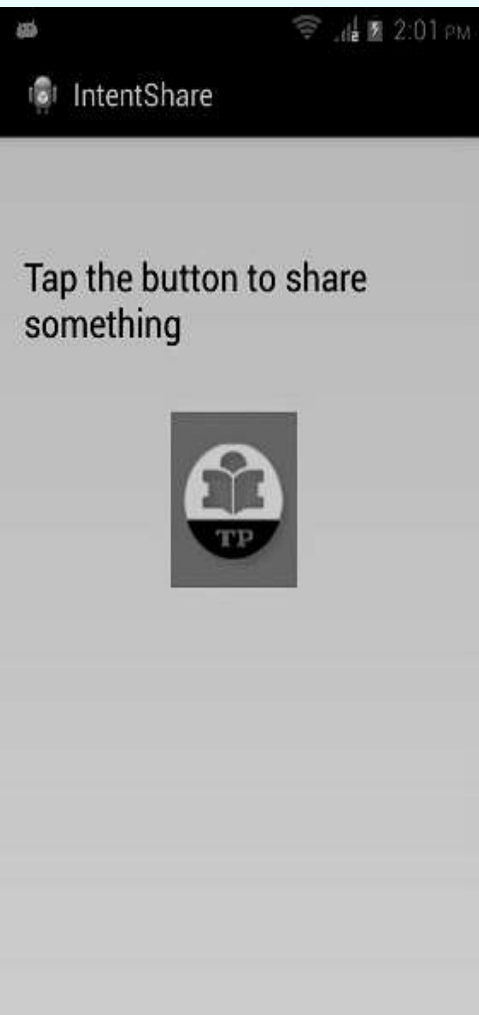
AndroidManifest.xml:

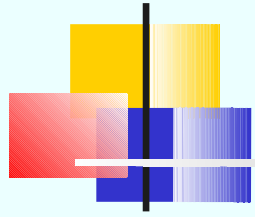
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.intentshare.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  < /activity>
</application>
```



Example (5)

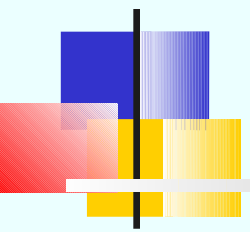




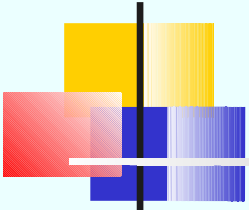
Exercises

From app of sudoku game or gumoku game:

- When a player wins, write a function to share the win screen on the facebook



Google maps



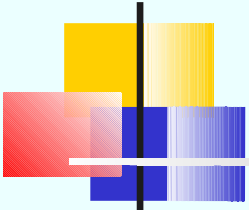
Google map (1)

Adding in layout file.xml:

```
<fragment  
    android:id="@+id/map"  
    android:name="com.google.android.gms.maps.MapFragment"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"/>
```

Adding in java code:

```
GoogleMap googleMap;  
googleMap = (MapFragment)  
    getSupportFragmentManager().findFragmentById(R.id.map).getMap();
```

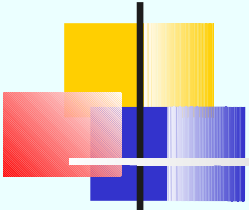


Google map (2)

Adding in manifest file.xml:

```
<uses-permission
android:name="android.permission.ACCESS_NETWORK_STATE"/>
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="com.google.android.providers.gsf.
    permission.READ_GSERVICES"/>
<uses-permission android:name="android.
    permission.WRITE_EXTERNAL_STORAGE"/>

<!--Google MAP API key-->
<meta-data
android:name="com.google.android.maps.v2.API_KEY"
android:value="AIzaSyDKymeBXNeiFWY5jRUejv6zItpmr2MVyQ0" />
```



Google map (3)

Adding marker:

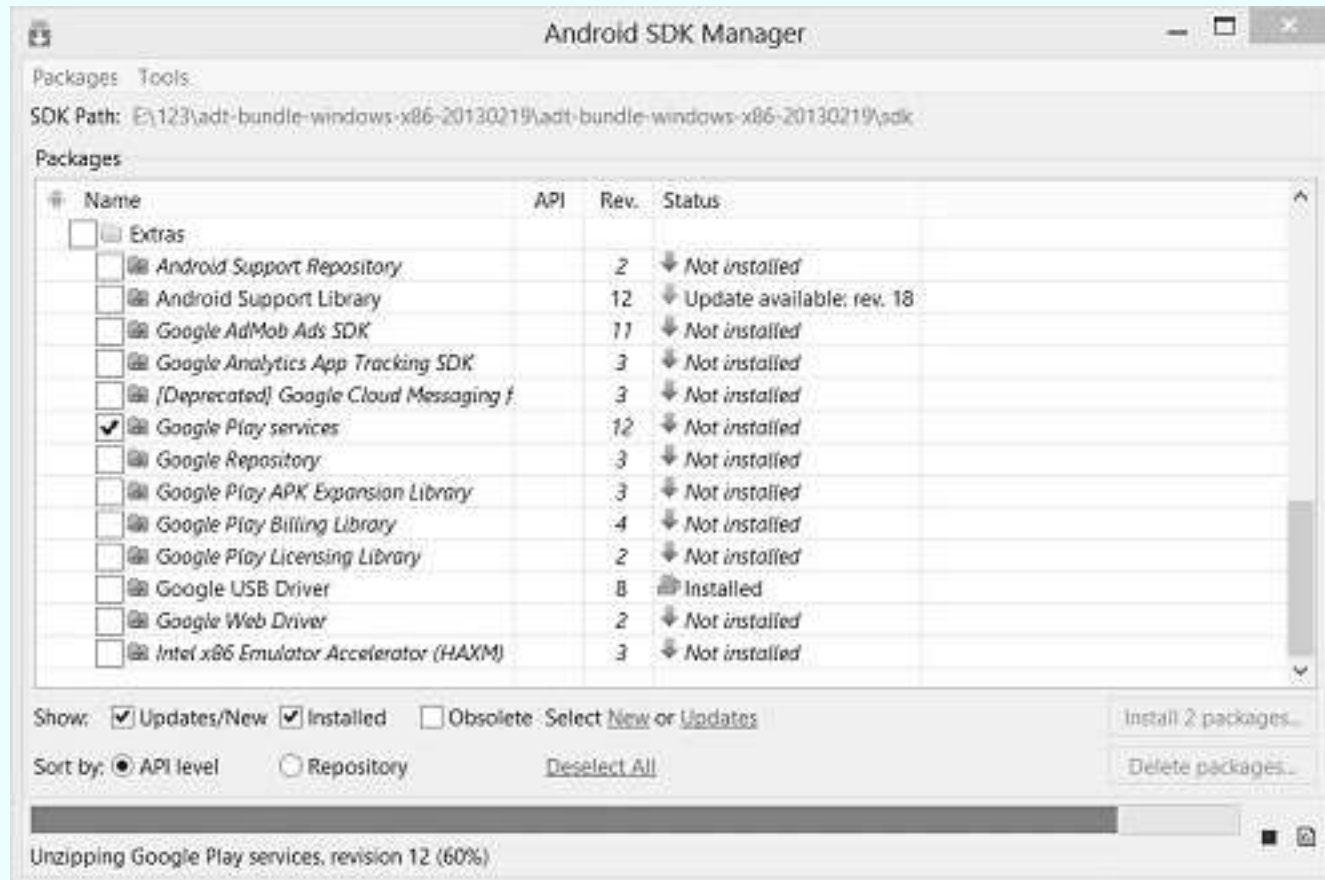
```
final LatLng TutorialPoint = new LatLng(21, 57);  
Marker TP = googleMap.addMarker(new  
MarkerOptions().position(TutorialPoint).title("TutorialPoint"));
```

Changing map type:

```
googleMap.setMapType(GoogleMap.MAP_TYPE_NORMAL);  
googleMap.setMapType(GoogleMap.MAP_TYPE_HYBRID);  
googleMap.setMapType(GoogleMap.MAP_TYPE_SATELLITE);  
googleMap.setMapType(GoogleMap.MAP_TYPE_TERRAIN);
```

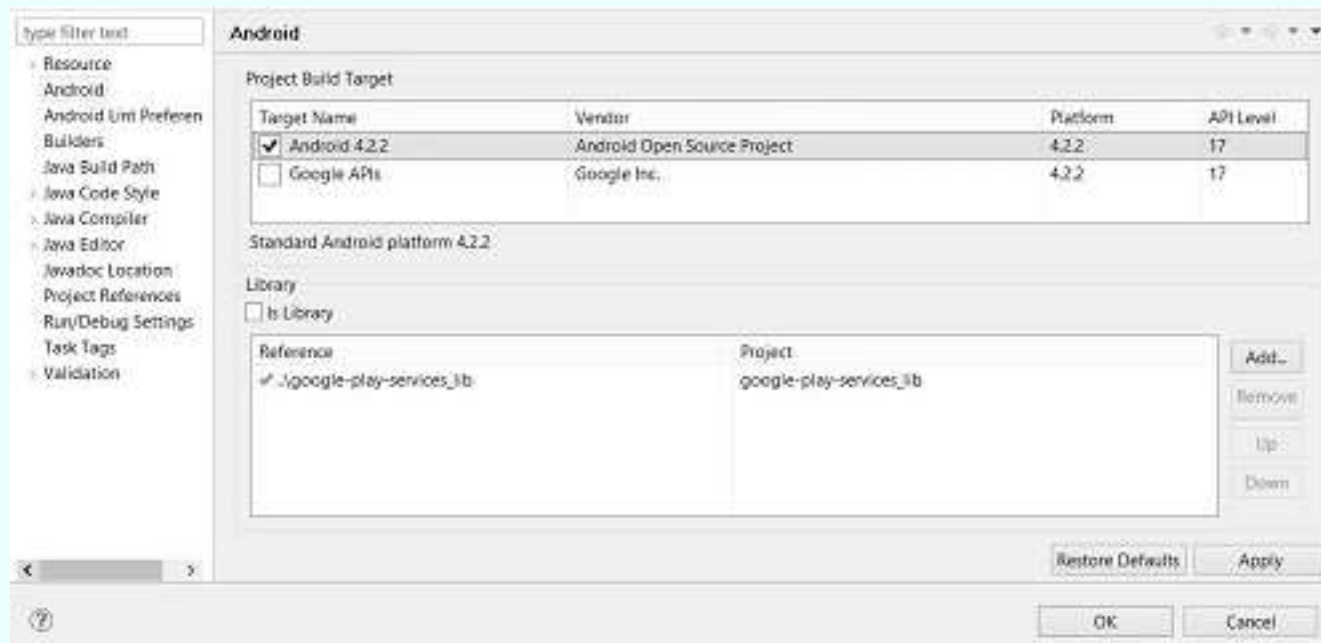
Integrating Google maps (1)

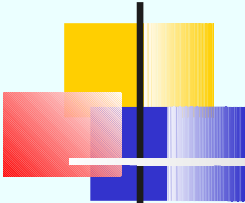
Install google service SDK and import into Eclipse:



Integrating Google maps (2)

Configure project with SDK:



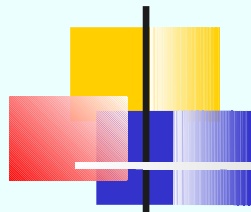


Integrating Google maps (3)

Generate certificate from KeyTool:

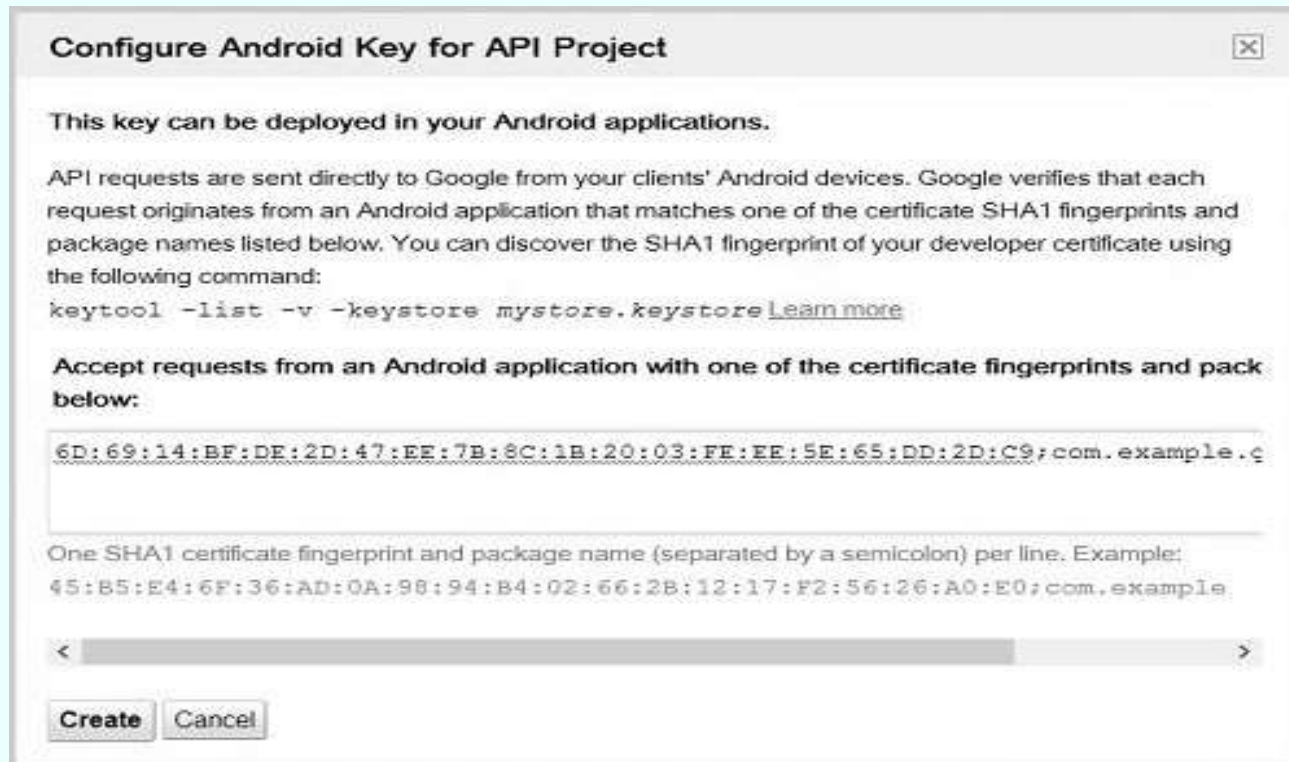
```
keytool -list -v -alias androiddebugkey -keystore %%Your path%% -  
storepass android -keypass android
```

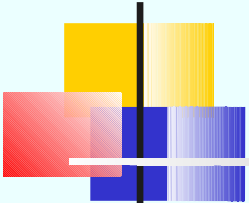
```
C:\Program Files (x86)\Java\jre7\bin>keytool -list -v -alias androiddebugkey -keystore C:\Users\...\android\debug.keystore -storepass android -keypass android  
Alias name: androiddebugkey  
Creation date: Oct 6, 2013  
Entry type: PrivateKeyEntry  
Certificate chain length: 1  
Certificate[1]:  
Owner: CN=Android Debug, O=Android, C=US  
Issuer: CN=Android Debug, O=Android, C=US  
Serial number: 352f55f0  
Valid from: ...  
Certificate fingerprints:  
MD5: 9D:49:75:15:1A:FC:1E:23:98:62:C8:29:72:EC:D2:71  
SHA1: AB:A9:D3:3E:92:D0:FE:4E:9D:21:87:8D:0D:61:18:8F:D8:91:E8:62  
SHA256: 14:8F:B7:40:BE:4B:83:2B:A9:13:CC:18:37:04:BB:12:45:F1:1E:B4:62:7D:3B:11:C9:AD:5E:CC:05:02:18:D9  
Signature algorithm name: SHA256withRSA  
Version: 3  
Extensions:
```



Integrating Google maps (4)

Getting key from Google console:

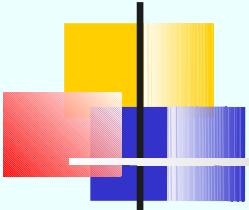




Integrating Google maps (5)

Specifying manifest:

```
<meta-data  
  android:name="com.google.android.maps.v2.API_KEY"  
  android:value="API_KEY"/>
```

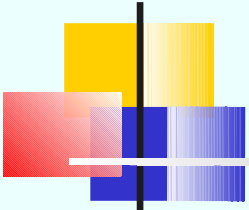


Example (1)

MainActivity.java:

```
public class MainActivity extends Activity {
    static final LatLng TutorialPoint = new LatLng(21, 57);
    private GoogleMap googleMap;

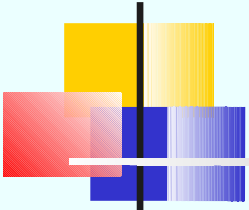
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        try {
            if (googleMap == null) {
                googleMap = ((MapFragment) getFragmentManager().
                    findFragmentById(R.id.map)).getMap();
            }
            googleMap.setMapType(GoogleMap.MAP_TYPE_HYBRID);
            Marker TP = googleMap.addMarker(new MarkerOptions().
                position(TutorialPoint).title("TutorialPoint"));
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```



Example (2)

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent" >
    <fragment
        android:id="@+id/map"
        android:name="com.google.android.gms.maps.MapFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</RelativeLayout>
```



Example (3)

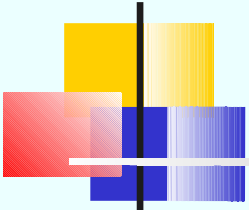
AndroidManifest.xml (1):

```
<uses-permission  
android:name="com.example.googlemaps.permission.MAPS_RECEIVE" />
```

```
<permission  
android:name="com.example.googlemaps.permission.MAPS_RECEIVE"  
android:protectionLevel="signature" />
```

```
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />  
<uses-permission android:name="android.permission.INTERNET" />  
<uses-permission android:name="com.google.android.providers.  
gsf.permission.READ_GSERVICES" />  
<uses-permission  
android:name="android.permission.WRITE_EXTERNAL_STORAGE" />  
<uses-permission  
android:name="android.permission.ACCESS_COARSE_LOCATION" />  
<uses-permission  
android:name="android.permission.ACCESS_FINE_LOCATION" />
```

```
<uses-feature  
android:glEsVersion="0x00020000"  
android:required="true" />
```



Example (4)

AndroidManifest.xml (2):

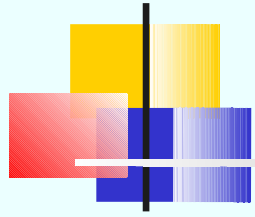
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.googlemaps.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>

  <meta-data
    android:name="com.google.android.maps.v2.API_KEY"
    android:value="AIzaSyDKymeBXNeiFWY5jRUejv6zItpmr2MVyQ0" />
</application>
```

Example (5)

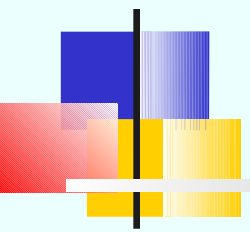




Exercises

Create an app:

- Recognize current position of user
- Enable user to share this info on the Facebook



Images

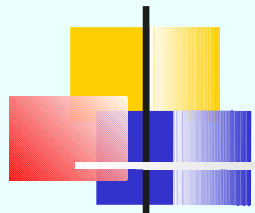


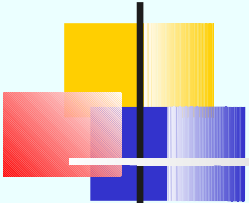
Image effects

Calling drawable of an image:

```
private Bitmap bmp;  
private ImageView img;  
img = (ImageView)findViewById(R.id.imageView1);  
BitmapDrawable abmp = (BitmapDrawable)img.getDrawable();
```

Accessing to a pixel of image:

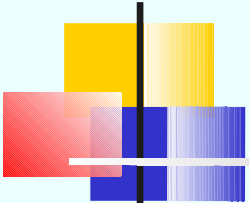
```
bmp = abmp.getBitmap();  
for(int i=0; i<bmp.getWidth(); i++){  
    for(int j=0; j<bmp.getHeight(); j++){  
        int p = bmp.getPixel(i, j);  
    }  
}
```



Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {
    private ImageView img;
    private Bitmap bmp;
    private Bitmap operation;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        img = (ImageView)findViewById(R.id.imageView1);
        BitmapDrawable abmp = (BitmapDrawable)img.getDrawable();
        bmp = abmp.getBitmap();
    }
    public void gray(View view){
        operation= Bitmap.createBitmap(bmp.getWidth(),
        bmp.getHeight(),bmp.getConfig());
        double red = 0.33; double green = 0.59; double blue = 0.11;
        for(int i=0; i<bmp.getWidth(); i++){
            for(int j=0; j<bmp.getHeight(); j++){
                int p = bmp.getPixel(i, j);
                int r = Color.red(p); int g = Color.green(p);
                int b = Color.blue(p);
                r = (int) red * r;      g = (int) green * g;
                b = (int) blue * b;
                operation.setPixel(i, j, Color.argb(
                    Color.alpha(p), r, g, b));
            }
        }
        img.setImageBitmap(operation);
    }
}
```

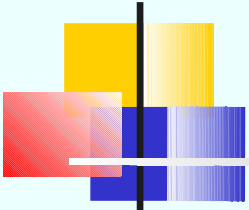


Example (2)

MainActivity.java (2):

```
public void bright(View view){
    operation= Bitmap.createBitmap bmp.getWidth(),
        bmp.getHeight(),bmp.getConfig());
    for(int i=0; i<bmp.getWidth(); i++){
        for(int j=0; j<bmp.getHeight(); j++){
            int p = bmp.getPixel(i, j); int r = Color.red(p);
            int g = Color.green(p); int b = Color.blue(p);
            int alpha = Color.alpha(p);
            r = 100 + r; g = 100 + g; b = 100 + b; alpha = 100 + alpha;
            operation.setPixel(i, j, Color.argb(alpha, r, g, b));
        }
    }
    img.setImageBitmap(operation);
}

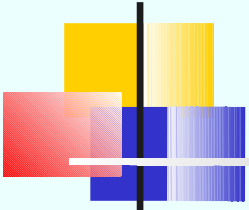
public void dark(View view){
    operation= Bitmap.createBitmap bmp.getWidth(),
        bmp.getHeight(),bmp.getConfig());
    for(int i=0; i<bmp.getWidth(); i++){
        for(int j=0; j<bmp.getHeight(); j++){
            int p = bmp.getPixel(i, j); int r = Color.red(p);
            int g = Color.green(p); int b = Color.blue(p);
            int alpha = Color.alpha(p); r = r - 50;
            g = g - 50; b = b - 50; alpha = alpha - 50;
            operation.setPixel(i, j, Color.argb(
                Color.alpha(p), r, g,b));
        }
    }
    img.setImageBitmap(operation);
}
```



Example (3)

activity_main.xml (1):

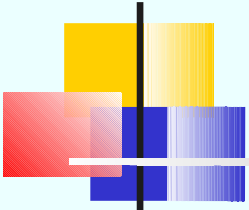
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >
    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBaseline="@+id/button1"
        android:layout_alignBottom="@+id/button1"
        android:layout_alignParentRight="true"
        android:layout_marginRight="19dp"
        android:onClick="dark"
        android:text="@string/dark" />
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_marginBottom="87dp"
        android:layout_marginRight="17dp"
        android:layout_toLeftOf="@+id/button3"
        android:onClick="gray"
        android:text="@string/gray" />
```



Example (4)

activity_main.xml (2):

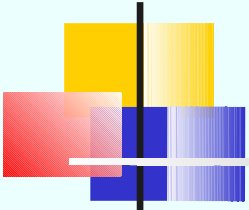
```
<Button
    android:id="@+id/button3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBaseline="@+id/button2"
    android:layout_alignBottom="@+id/button2"
    android:layout_centerHorizontal="true"
    android:onClick="bright"
    android:text="@string/bright" />
<ImageView
    android:id="@+id/imageView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentTop="true"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="114dp"
    android:src="@drawable/ic_launcher" />
</RelativeLayout>
```



Example (5)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <string name="app_name">ImageEffects</string>
  <string name="action_settings">Settings</string>
  <string name="hello_world">Hello world!</string>
  <string name="gray">Gray</string>
  <string name="bright">bright</string>
  <string name="dark">dark</string>
</resources>
```



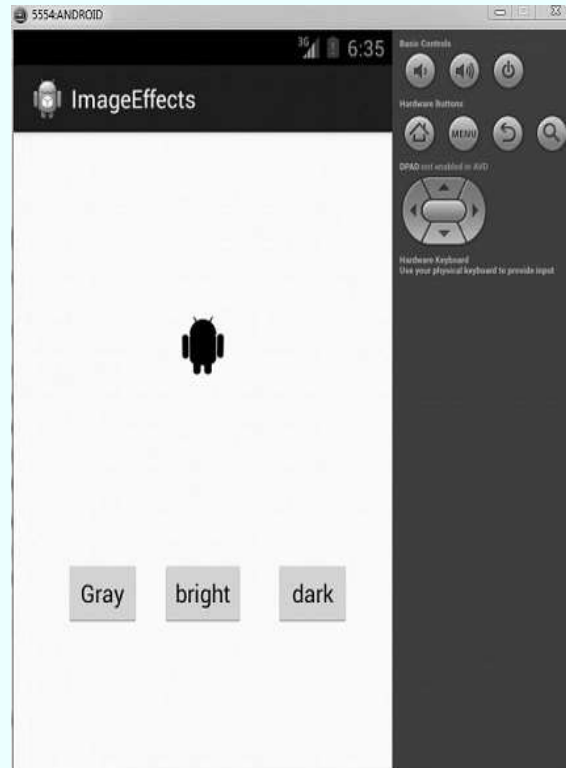
Example (6)

AndroidManifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.imageeffects.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (7)



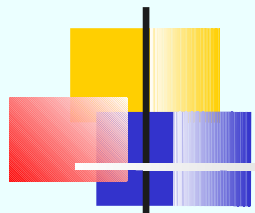


Image switcher

Create:

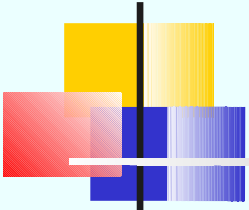
```
private ImageSwitcher imageSwitcher;  
imageSwitcher = (ImageSwitcher)findViewById(R.id.imageSwitcher1);
```

Config:

```
imageSwitcher.setImageResource(R.drawable.ic_launcher);  
imageSwitcher.setFactory(new ViewFactory() {  
    public View makeView() {  
        ImageView myView = new ImageView(getApplicationContext());  
        return myView;  
    }  
})
```

Animation:

```
Animation in =  
    AnimationUtils.loadAnimation(this, android.R.anim.slide_in_left);  
imageSwitcher.setInAnimation(in);  
imageSwitcher.setOutAnimation(out);
```

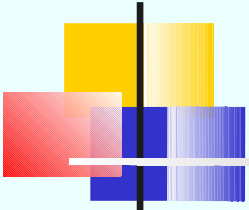


Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {
    private ImageButton img;
    private ImageSwitcher imageSwitcher;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        img = (ImageButton)findViewById(R.id.imageButton1);
        imageSwitcher = (ImageSwitcher)findViewById(R.id.imageSwitcher);
        imageSwitcher.setFactory(new ViewFactory() {
            @Override
            public View makeView() {
                ImageView myView = new ImageView(getApplicationContext());
                myView.setScaleType(ImageView.ScaleType.FIT_CENTER);
                myView.setLayoutParams(new ImageSwitcher.LayoutParams(
                    LayoutParams.FILL_PARENT, LayoutParams.FILL_PARENT));
                return myView;
            }
        });
    }
}
```

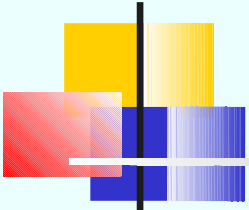


Example (2)

MainActivity.java (2):

```
public void next(View view) {
    Toast.makeText(getApplicationContext(), "Next Image",
        Toast.LENGTH_LONG).show();
    Animation in = AnimationUtils.loadAnimation(this,
        android.R.anim.slide_in_left);
    Animation out = AnimationUtils.loadAnimation(this,
        android.R.anim.slide_out_right);
    imageSwitcher.setInAnimation(in);
    imageSwitcher.setOutAnimation(out);
    imageSwitcher.setImageResource(R.drawable.ic_launcher);
}

public void previous(View view) {
    Toast.makeText(getApplicationContext(), "previous Image",
        Toast.LENGTH_LONG).show();
    Animation in = AnimationUtils.loadAnimation(this,
        android.R.anim.slide_out_right);
    Animation out = AnimationUtils.loadAnimation(this,
        android.R.anim.slide_in_left);
    imageSwitcher.setInAnimation(out);
    imageSwitcher.setOutAnimation(in);
    imageSwitcher.setImageResource(R.drawable.ic_launcher);
}
```



Example (3)

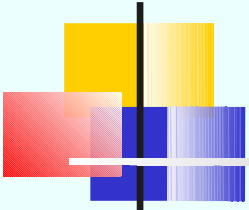
activity_main.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity" >
```

```
<ImageButton
android:id="@+id/imageButton1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="54dp"
android:onClick="next"
android:src="@android:drawable/ic_menu_send" />
```

```
<ImageSwitcher
android:id="@+id/imageSwitcher1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_centerHorizontal="true"
android:layout_centerVertical="true" >
</ImageSwitcher>
```

```
<ImageButton
android:id="@+id/imageButton2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentBottom="true"
android:layout_centerHorizontal="true"
android:layout_marginBottom="85dp"
android:onClick="previous"
android:src="@android:drawable/ic_menu_revert" />
```



Example (4)

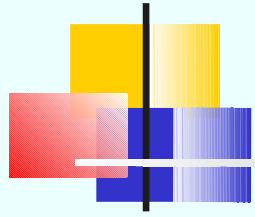
Manifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.imageswitcher.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (5)

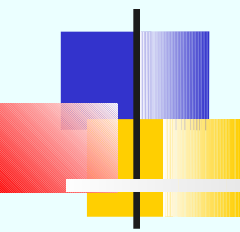




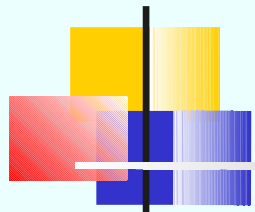
Exercises

Create an app:

- Enable to read an image from phone to edit
- Enable user to share this photo on the Facebook



Internal storage



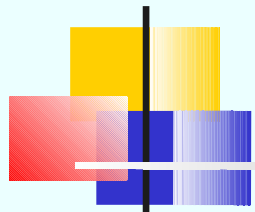
Writing file

Create:

```
FileOutputStream fOut = openFileOutput("file name here",  
    MODE_WORLD_READABLE);
```

Using:

```
String str = "data";  
fOut.write(str.getBytes());  
fOut.close();
```



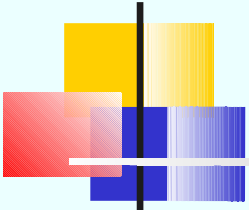
Reading file

Create:

```
FileInputStream fin = openFileInput(file);
```

Using:

```
int c;  
String temp="";  
while( (c = fin.read()) != -1){  
    temp = temp + Character.toString((char)c);  
}  
//string temp contains all the data of the file.  
fin.close();
```



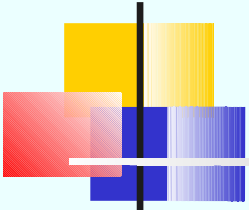
Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {
    private EditText et;
    private String data;
    private String file = "mydata";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        et = (EditText) (findViewById(R.id.editText1));
    }

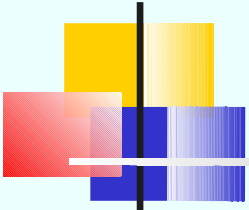
    public void save(View view) {
        data = et.getText().toString();
        try {
            FileOutputStream fOut =
                openFileOutput(file, MODE_WORLD_READABLE);
            fOut.write(data.getBytes());
            fOut.close();
            Toast.makeText(getApplicationContext(), "file saved",
                Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```



Example (2)

MainActivity.java (2):

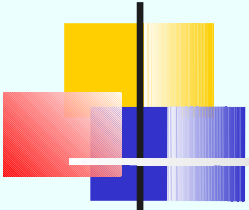
```
public void read(View view) {
    try{
        FileInputStream fin = openFileInput(file);
        int c;
        String temp="";
        while( (c = fin.read()) != -1){
            temp = temp + Character.toString((char)c);
        }
        et.setText(temp);
        Toast.makeText(getApplicationContext(),"file read",
            Toast.LENGTH_SHORT).show();
    }catch(Exception e){
    }
}
```



Example (3)

activity_main.xml:

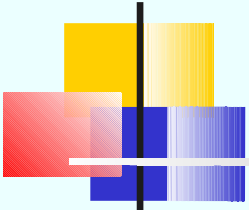
```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity" >
<Button
android:id="@+id/button1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="182dp"
android:onClick="save"
android:text="@string/save" />
<Button
android:id="@+id/button2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/button1"
android:layout_alignRight="@+id/button1"
android:layout_below="@+id/button1"
android:layout_marginTop="46dp"
android:onClick="read"
android:text="@string/read" />
<EditText
android:id="@+id/editText1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/button1"
android:layout_alignParentTop="true"
android:layout_marginTop="23dp"
android:ems="10"
android:inputType="textMultiLine" />
```



Example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <string name="app_name">Storage</string>
  <string name="action_settings">Settings</string>
  <string name="hello_world">Hello world!</string>
  <string name="save">save to internal storage</string>
  <string name="read">load from internal storage</string>
</resources>
```



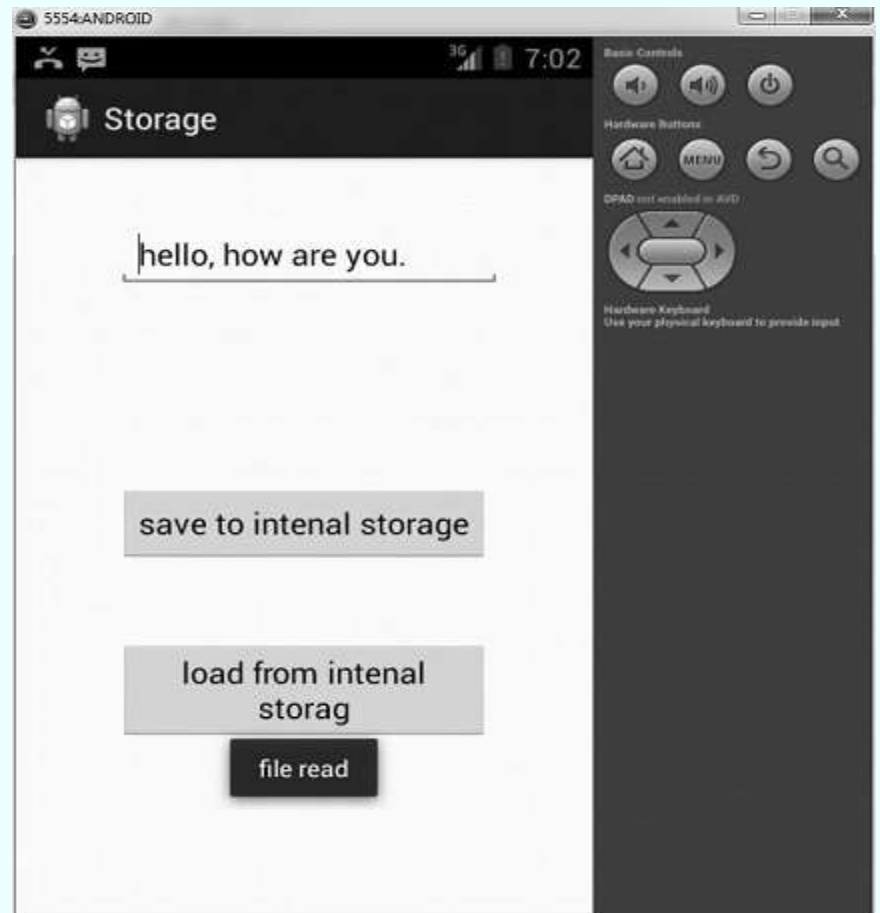
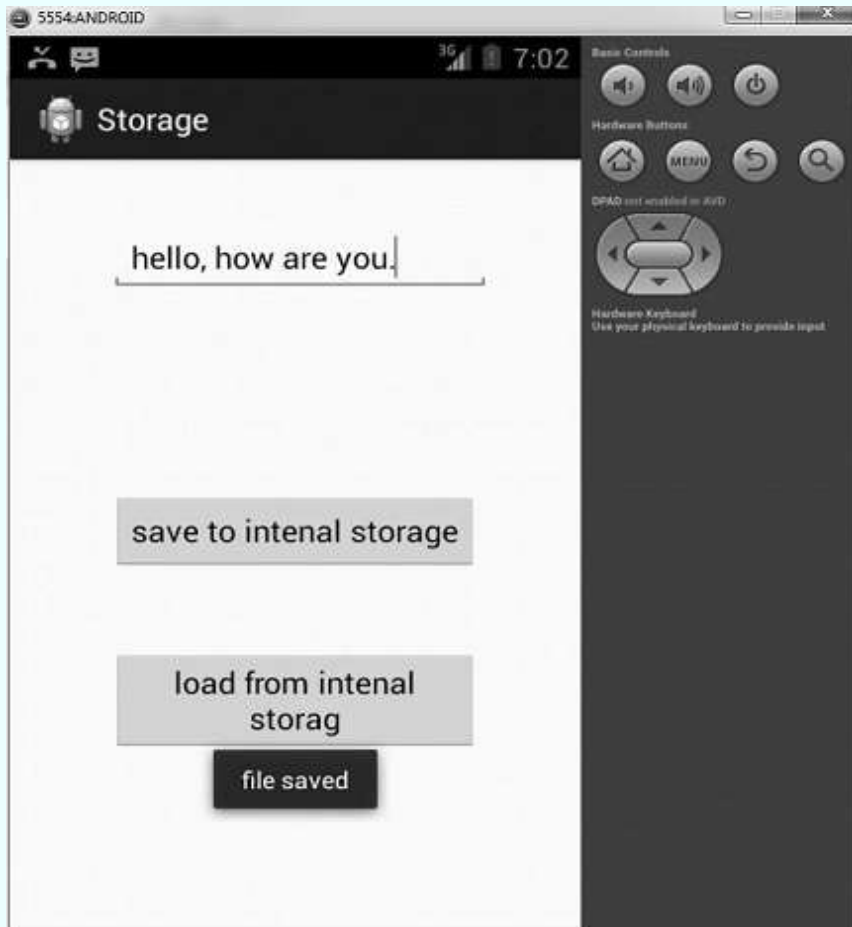
Example (5)

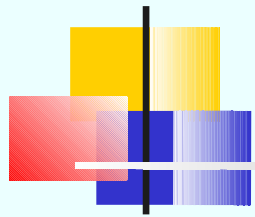
AndroidManifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.storage.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (6)

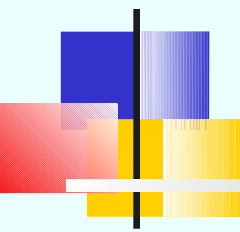




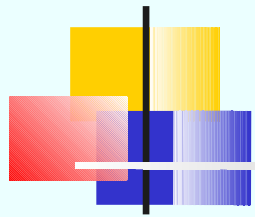
Exercises

Create an app:

- Enable to read an folder to list all files
- Enable to select some files to copy
- Enable to paste files to another folder



JSON parser



JSONObject

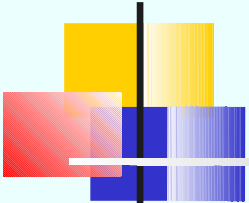
create:

```
String in; // source of JSON
JSONObject reader = new JSONObject(in);
```

Using:

```
JSONObject sys = reader.getJSONObject("sys");
country = sys.getString("country");
JSONObject main = reader.getJSONObject("main");
temperature = main.getString("temp");
```

```
{
  "sys":
    {
      "country": "GB",
      "sunrise": 1381107633,
      "sunset": 1381149604
    },
  "weather": [
    {
      "id": 711,
      "main": "Smoke",
      "description": "smoke",
      "icon": "50n"
    }
  ],
  "main":
    {
      "temp": 304.15,
      "pressure": 1009,
    }
}
```



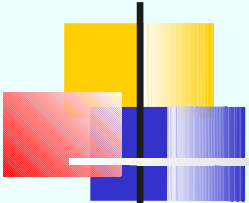
Example (1)

MainActivity.java:

```
public class MainActivity extends Activity {
    private String url1 =
        "http://api.openweathermap.org/data/2.5/weather?q=";
    private EditText location, country, temperature, humidity, pressure;
    private HandleJSON obj;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        location = (EditText)findViewById(R.id.editText1);
        country = (EditText)findViewById(R.id.editText2);
        temperature = (EditText)findViewById(R.id.editText3);
        humidity = (EditText)findViewById(R.id.editText4);
        pressure = (EditText)findViewById(R.id.editText5);
    }

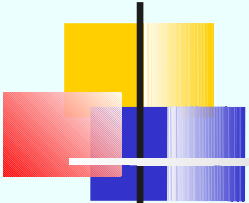
    public void open(View view) {
        String url = location.getText().toString();
        String finalUrl = url1 + url;
        country.setText(finalUrl);
        obj = new HandleJSON(finalUrl);
        obj.fetchJSON();
        while(obj.parsingComplete);
        country.setText(obj.getCountry());
        temperature.setText(obj.getTemperature());
        humidity.setText(obj.getHumidity());
        pressure.setText(obj.getPressure());
    }
}
```



Example (2)

HandleXML.java (1):

```
public class HandleJSON {
    private String country = "county";
    private String temperature = "temperature";
    private String humidity = "humidity";
    private String pressure = "pressure";
    private String urlString = null;
    public volatile boolean parsingComplete = true;
    public HandleJSON(String url){
        this.urlString = url;
    }
    public String getCountry(){
        return country;
    }
    public String getTemperature(){
        return temperature;
    }
    public String getHumidity(){
        return humidity;
    }
    public String getPressure(){
        return pressure;
    }
    static String convertStreamToString(java.io.InputStream is) {
        java.util.Scanner s = new
            java.util.Scanner(is).useDelimiter("\\A");
        return s.hasNext() ? s.next() : "";
    }
}
```

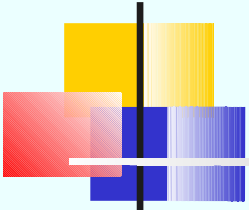


Example (3)

HandleXML.java (2):

```
@SuppressWarnings("NewApi")
public void readAndParseJSON(String in) {
    try {
        JSONObject reader = new JSONObject(in);
        JSONObject sys = reader.getJSONObject("sys");
        country = sys.getString("country");
        JSONObject main = reader.getJSONObject("main");
        temperature = main.getString("temp");
        pressure = main.getString("pressure");
        humidity = main.getString("humidity");
        parsingComplete = false;
    } catch (Exception e) {e.printStackTrace();}
}

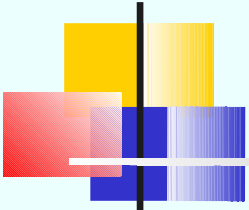
public void fetchJSON(){
    Thread thread = new Thread(new Runnable(){
        @Override
        public void run() {
            try {
                URL url = new URL(urlString);
                HttpURLConnection conn = (HttpURLConnection)
                    url.openConnection();
                conn.setReadTimeout(10000 /* milliseconds */);
                conn.setConnectTimeout(15000 /* milliseconds */);
                conn.setRequestMethod("GET");
                conn.setDoInput(true);
                conn.connect();
                InputStream stream = conn.getInputStream();
                String data = convertStreamToString(stream);
                readAndParseJSON(data);
                stream.close();
            } catch (Exception e) {e.printStackTrace();}
        }
    });
    thread.start();
}
```



Example (4)

activity_main.xml (1):

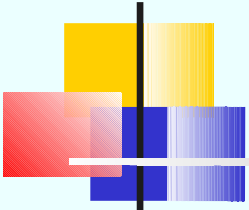
```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
tools:context=".MainActivity" >
<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentLeft="true"
android:layout_alignParentTop="true"
android:layout_marginTop="15dp"
android:text="@string/location"
android:textAppearance="?android:attr/textAppearanceMedium" />
<EditText
android:id="@+id/editText1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBottom="@+id/textView1"
android:layout_alignParentRight="true"
android:ems="10" />
<TextView
android:id="@+id/textView2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/textView1"
android:layout_below="@+id/textView1"
android:layout_marginTop="68dp"
android:text="@string/country"
android:textAppearance="?android:attr/textAppearanceSmall" />
```



Example (5)

activity_main.xml (2):

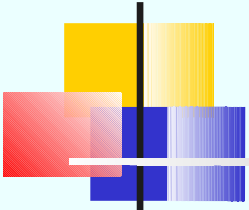
```
<TextView
android:id="@+id/textView3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@+id/textView2"
android:layout_marginTop="19dp"
android:text="@string/temperature"
android:textAppearance="?android:attr/textAppearanceSmall" />
<TextView
android:id="@+id/textView4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/textView3"
android:layout_below="@+id/textView3"
android:layout_marginTop="32dp"
android:text="@string/humidity"
android:textAppearance="?android:attr/textAppearanceSmall" />
<TextView
android:id="@+id/textView5"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/textView4"
android:layout_below="@+id/textView4"
android:layout_marginTop="21dp"
android:text="@string/pressure"
android:textAppearance="?android:attr/textAppearanceSmall" />
<EditText
android:id="@+id/editText2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_above="@+id/textView3"
android:layout_toRightOf="@+id/textView3"
android:ems="10" >
</EditText>
</requestFocus />
```



Example (6)

activity_main.xml (3):

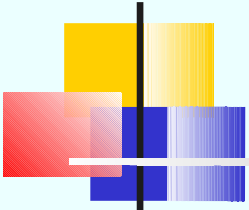
```
<EditText
android:id="@+id/editText3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/textView3"
android:layout_alignBottom="@+id/textView3"
android:layout_alignLeft="@+id/editText2"
android:ems="10" />
<EditText
android:id="@+id/editText4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_above="@+id/textView5"
android:layout_alignLeft="@+id/editText1"
android:ems="10" />
<EditText
android:id="@+id/editText5"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/textView5"
android:layout_alignBottom="@+id/textView5"
android:layout_alignRight="@+id/editText4"
android:ems="10" />
<Button
android:id="@+id/button1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignLeft="@+id/editText2"
android:layout_below="@+id/editText1"
android:onClick="open"
android:text="@string/weather" />
</RelativeLayout>
```



Example (7)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <string name="app_name">JSONParser</string>
  <string name="action_settings">Settings</string>
  <string name="hello_world">Hello world!</string>
  <string name="location">Location</string>
  <string name="country">Country:</string>
  <string name="temperature">Temperature:</string>
  <string name="humidity">Humidity:</string>
  <string name="pressure">Pressure:</string>
  <string name="weather">Weather</string>
</resources>
```



Example (8)

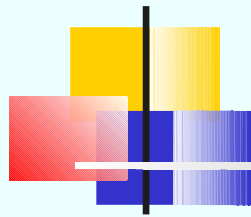
AndroidManifest.xml:

```
<uses-permission android:name="android.permission.INTERNET" />

<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.jsonparser.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Example (9)

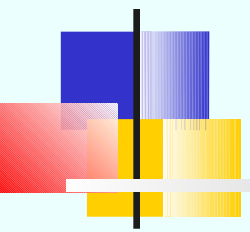




Exercises

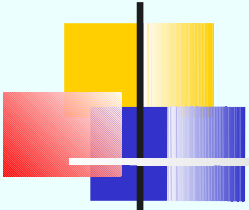
Create an app:

- Enable to fill automatically a keyword to google.com
- Enable to read return results to parser to get list of results object to display



Loading spinner





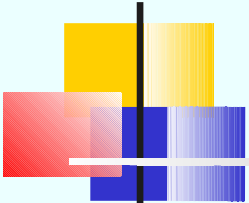
ProgressBar Object

Create in xml layout:

```
<ProgressBar  
    android:id="@+id/progressBar1"  
    style="?android:attr/progressBarStyleLarge"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_centerHorizontal="true" />
```

Using in Java code:

```
private ProgressBar spinner;  
spinner = (ProgressBar)findViewById(R.id.progressBar1);  
spinner.setVisibility(View.GONE);  
spinner.setVisibility(View.VISIBLE);
```



Example (1)

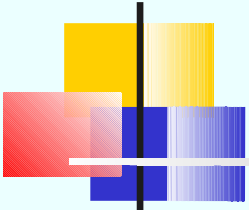
MainActivity.java:

```
public class MainActivity extends Activity {
    private ProgressBar spinner;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        spinner = (ProgressBar)findViewById(R.id.progressBar1);
        spinner.setVisibility(View.GONE);
    }

    public void load(View view){
        spinner.setVisibility(View.VISIBLE);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }
}
```



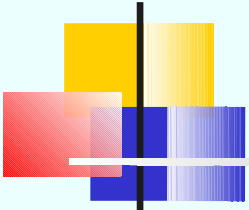
Example (2)

activity_main.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
<Button
android:id="@+id/button1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="103dp"
android:onClick="load"
android:text="@string/hello_world" />
```

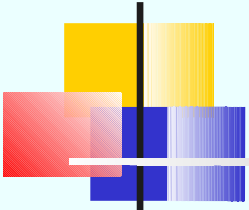
```
<ProgressBar
android:id="@+id/progressBar1"
style="?android:attr/progressBarStyleLarge"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@+id/button1"
android:layout_centerHorizontal="true" />
</RelativeLayout>
```



Example (3)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">Spinner</string>
<string name="action_settings">Settings</string>
<string name="hello_world">load spinner</string>
</resources>
```

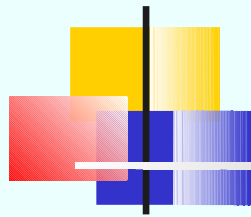


Example (4)

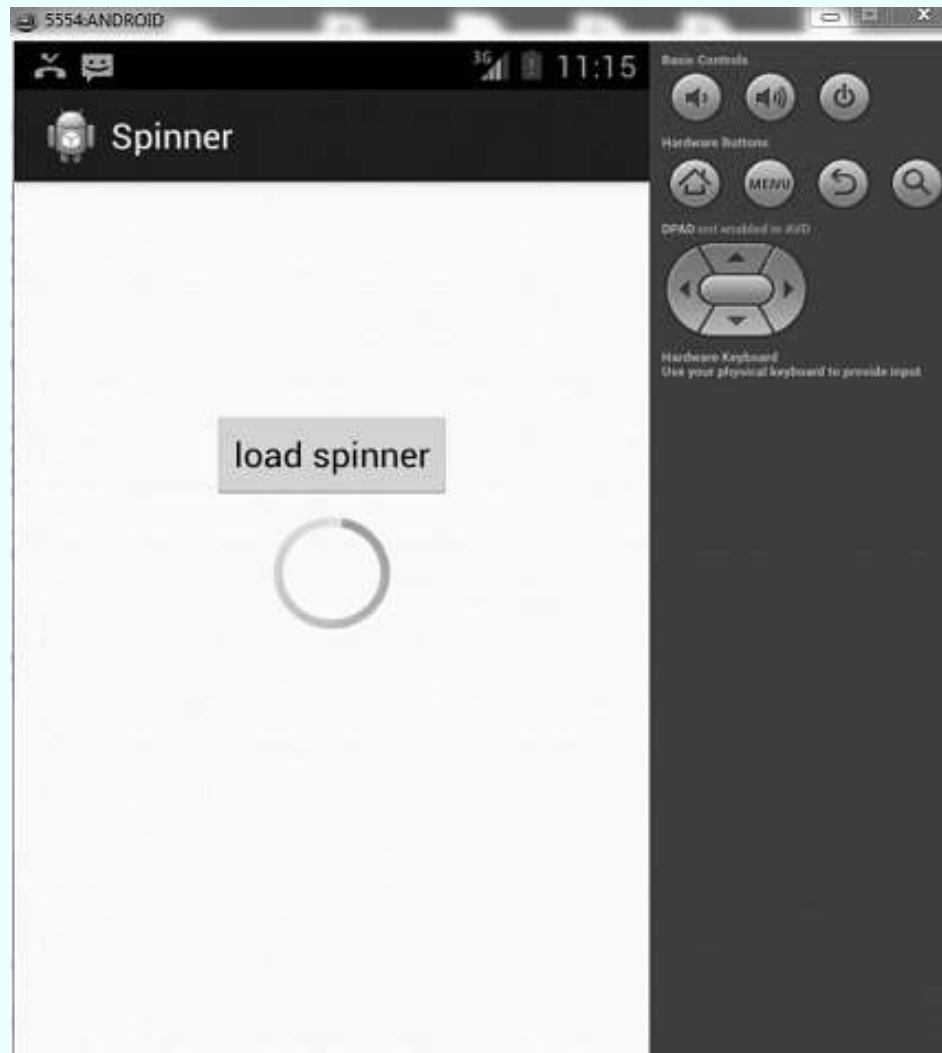
AndroidManifest.xml:

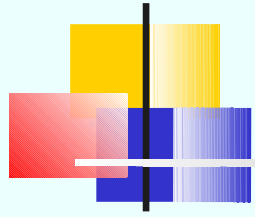
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.spinner.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```



Example (5)

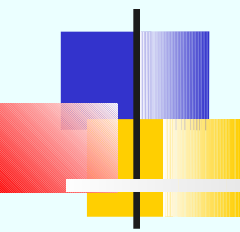




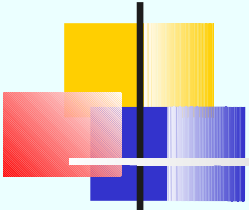
Exercises

For the app copying file:

- Add loading spinner into the app



Localization



Localizing strings (1)

res/values/strings.xml (en - default):

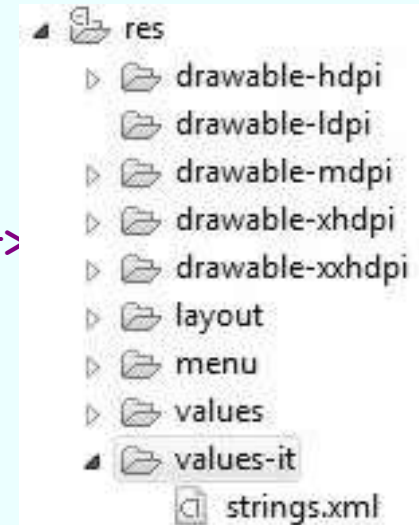
```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="hello_world">Hello World!</string>
</resources>
```

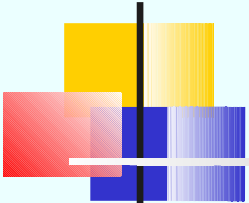
res/values-fr/strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="hello_world">Bonjour le monde !</string>
</resources>
```

res/values-it/strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="hello_world">Ciao mondo!</string>
</resources>
```





Localizing strings (2)

Language strings:

-en: default

-fr: French

-es: Espagnol

-pt: portuguese

-it: italian

-de: german

-ru: russian

-da: danish

-bg: bulgarian

-sv: swedish

-ur: ukrainian

-tr: turkish

-jp: japanese

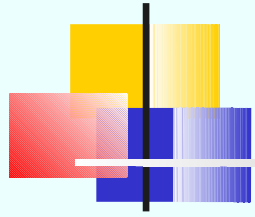
-th: thai

-hi: hindi

-ko: korean

-zh: chinesse

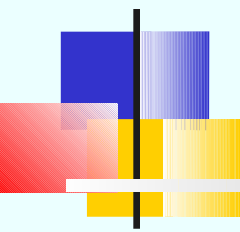
-vi: vietnamese



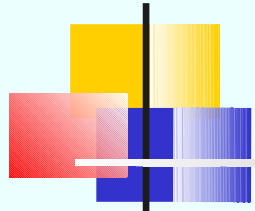
Exercises

For the app created:

- Add local language into all apps



Media player



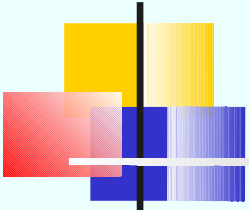
MediaPlayer object

Create:

```
MediaPlayer mediaPlayer = MediaPlayer.create(this, R.raw.song);
```

Using:

```
mediaPlayer.start();  
mediaPlayer.pause();  
mediaPlayer.reset();
```

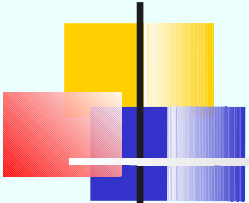


Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {
    public TextView songName,startTimeField,endTimeField;
    private MediaPlayer mediaPlayer;
    private double startTime = 0;
    private double finalTime = 0;
    private Handler myHandler = new Handler();
    private int forwardTime = 5000;
    private int backwardTime = 5000;
    private SeekBar seekbar;
    private ImageButton playButton,pauseButton;
    public static int oneTimeOnly = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        songName = (TextView)findViewById(R.id.textView4);
        startTimeField =(TextView)findViewById(R.id.textView1);
        endTimeField =(TextView)findViewById(R.id.textView2);
        seekbar = (SeekBar)findViewById(R.id.seekBar1);
        playButton = (ImageButton)findViewById(R.id.imageButton1);
        pauseButton = (ImageButton)findViewById(R.id.imageButton2);
        songName.setText("song.mp3");
        mediaPlayer = MediaPlayer.create(this, R.raw.song);
        seekbar.setClickable(false);
        pauseButton.setEnabled(false);
    }
}
```

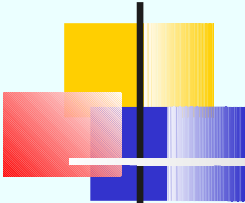


Example (2)

MainActivity.java (2):

```
public void play(View view) {
    Toast.makeText(getApplicationContext(), "Playing sound",
        Toast.LENGTH_SHORT).show();
    mediaPlayer.start();
    finalTime = mediaPlayer.getDuration();
    startTime = mediaPlayer.getCurrentPosition();
    if (oneTimeOnly == 0) {
        seekbar.setMax((int) finalTime);
        oneTimeOnly = 1;
    }
    endTimeField.setText(String.format("%d min, %d sec",
        TimeUnit.MILLISECONDS.toMinutes((long) finalTime),
        TimeUnit.MILLISECONDS.toSeconds((long) finalTime) -
        TimeUnit.MINUTES.toSeconds(TimeUnit.MILLISECONDS.
            toMinutes((long) finalTime))));
    startTimeField.setText(String.format("%d min, %d sec",
        TimeUnit.MILLISECONDS.toMinutes((long) startTime),
        TimeUnit.MILLISECONDS.toSeconds((long) startTime) -
        TimeUnit.MINUTES.toSeconds(TimeUnit.MILLISECONDS.
            toMinutes((long) startTime))));
    seekbar.setProgress((int) startTime);
    myHandler.postDelayed(UpdateSongTime, 100);
    pauseButton.setEnabled(true);
    playButton.setEnabled(false);
}

private Runnable UpdateSongTime = new Runnable() {
    public void run() {
        startTime = mediaPlayer.getCurrentPosition();
        startTimeField.setText(String.format("%d min, %d sec",
            TimeUnit.MILLISECONDS.toMinutes((long) startTime),
            TimeUnit.MILLISECONDS.toSeconds((long) startTime) -
            TimeUnit.MINUTES.toSeconds(TimeUnit.MILLISECONDS.
                toMinutes((long) startTime))));
        seekbar.setProgress((int) startTime);
        myHandler.postDelayed(this, 100);
    }
}
```



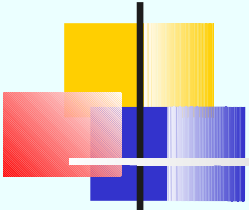
Example (3)

MainActivity.java (3):

```
public void pause(View view){
    Toast.makeText(getApplicationContext(), "Pausing sound",
        Toast.LENGTH_SHORT).show();
    mediaPlayer.pause();
    pauseButton.setEnabled(false);
    playButton.setEnabled(true);
}

public void forward(View view){
    int temp = (int)startTime;
    if((temp+forwardTime)<=finalTime){
        startTime = startTime + forwardTime;
        mediaPlayer.seekTo((int) startTime);
    }else{
        Toast.makeText(getApplicationContext(),
            "Cannot jump forward 5 seconds",Toast.LENGTH_SHORT).show()
    }
}

public void rewind(View view){
    int temp = (int)startTime;
    if((temp-backwardTime)>0){
        startTime = startTime - backwardTime;
        mediaPlayer.seekTo((int) startTime);
    }else{
        Toast.makeText(getApplicationContext(),
            "Cannot jump backward 5 seconds",Toast.LENGTH_SHORT).show()
    }
}
```



Example (4)

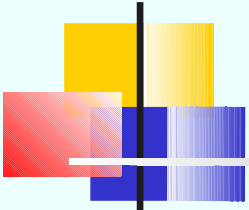
activity_main.xml (1):

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" >
```

```
<ImageButton
    android:id="@+id/imageButton3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentBottom="true"
    android:layout_alignParentLeft="true"
    android:layout_marginBottom="14dp"
    android:onClick="forward"
    android:src="@android:drawable/ic_media_ff" />
```

```
<ImageButton
    android:id="@+id/imageButton4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentRight="true"
    android:layout_alignTop="@+id/imageButton2"
    android:layout_marginLeft="22dp"
    android:layout_toRightOf="@+id/imageButton2"
    android:onClick="rewind"
    android:src="@android:drawable/ic_media_rew" />
```

```
<ImageButton
    android:id="@+id/imageButton2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignTop="@+id/imageButton1"
    android:layout_marginLeft="14dp"
    android:layout_toRightOf="@+id/imageButton1"
    android:onClick="pause"
```



Example (5)

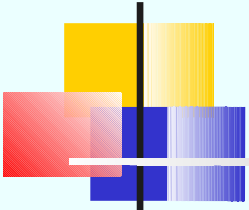
activity_main.xml (2):

```
<ImageButton
    android:id="@+id/imageButton1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignTop="@+id/imageButton3"
    android:layout_marginLeft="24dp"
    android:layout_toRightOf="@+id/imageButton3"
    android:onClick="play"
    android:src="@android:drawable/ic_media_play" />

<SeekBar
    android:id="@+id/seekBar1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_above="@+id/imageButton3"
    android:layout_toLeftOf="@+id/textView2"
    android:layout_toRightOf="@+id/textView1" />

<TextView
    android:id="@+id/textView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignRight="@+id/imageButton3"
    android:layout_alignTop="@+id/seekBar1"
    android:text="@string/initial_time"
    android:textAppearance="?android:attr/textAppearanceSmall" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/imageButton4"
    android:layout_alignTop="@+id/seekBar1"
    android:text="@string/initial_time"
    android:textAppearance="?android:attr/textAppearanceSmall" />
```



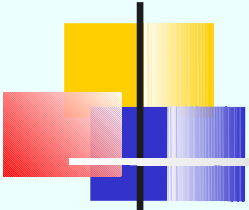
Example (6)

activity_main.xml (3):

```
<TextView
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/imageButton3"
    android:text="@string/hello_world"
    android:textAppearance="?android:attr/textAppearanceMedium" />
```

```
<ImageView
    android:id="@+id/imageView1"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_alignParentBottom="true"
    android:layout_alignParentLeft="true"
    android:layout_below="@+id/textView3"
    android:src="@drawable/ic_launcher" />
```

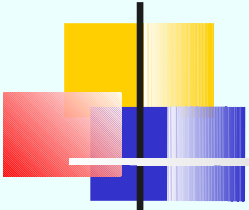
```
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBaseline="@+id/textView3"
    android:layout_alignBottom="@+id/textView3"
    android:layout_toRightOf="@+id/imageButton1"
    android:text="TextView" />
</RelativeLayout>
```



Example (7)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">MediaPlayer</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Now Playing:</string>
<string name="inital_Time">0 min, 0 sec</string>
</resources>
```

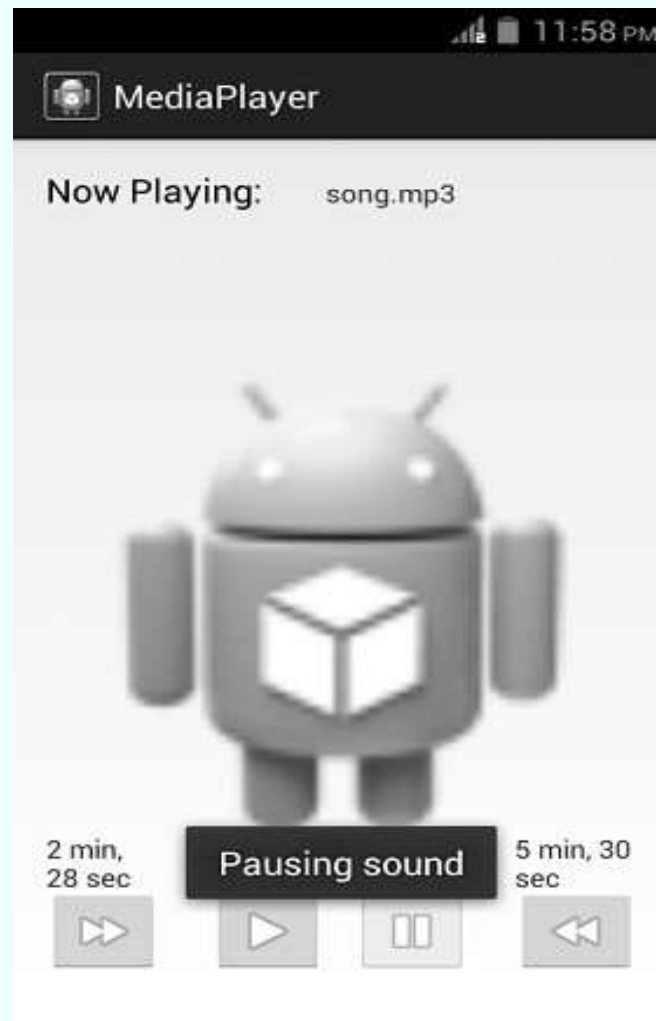
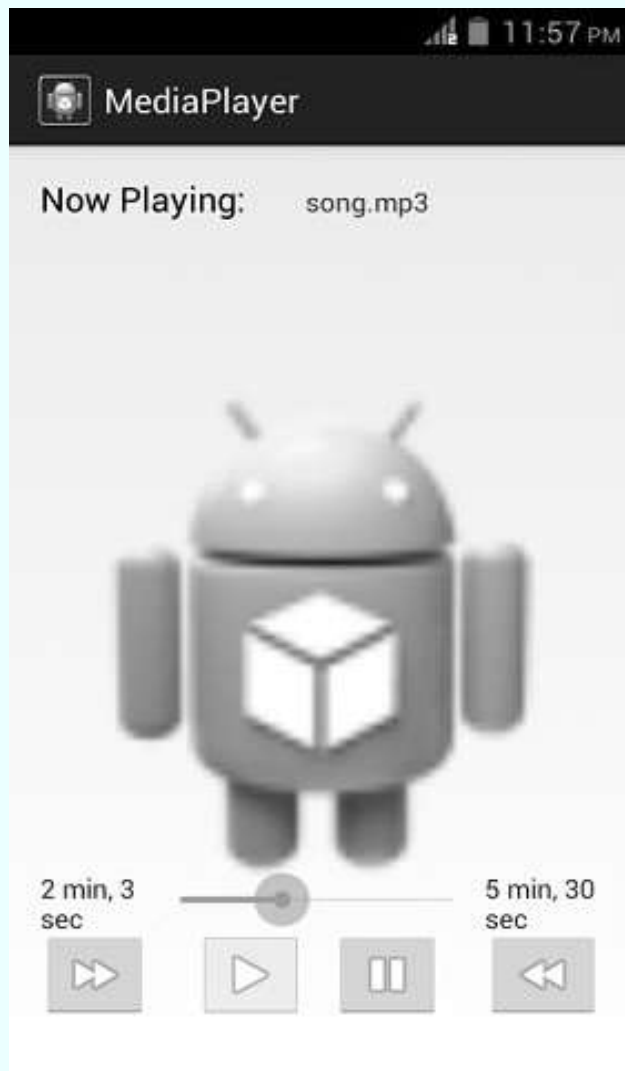


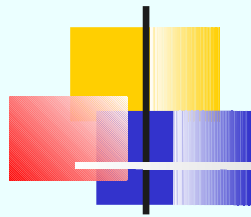
Example (8)

manifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >
  <activity
    android:name="com.example.mediaplayer.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (9)

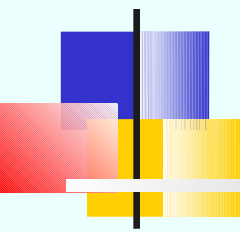




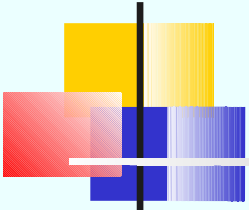
Exercises

Create an app:

- Enable to read music file from internal storage to add to an album
- Select an album to play



Network connection



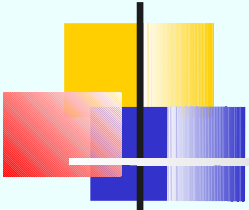
Checking connection

Create:

```
ConnectivityManager check = (ConnectivityManager)
    this.context.getSystemService(Context.CONNECTIVITY_SERVICE);
```

Using:

```
NetworkInfo[] info = check.getAllNetworkInfo();
for (int i = 0; i<info.length; i++){
    if (info[i].getState() == NetworkInfo.State.CONNECTED) {
        Toast.makeText(context, "Internet is connected",
            Toast.LENGTH_SHORT).show();
    }
}
```



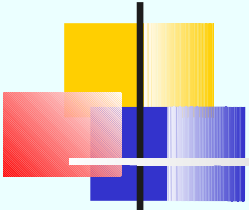
Connect

Create:

```
String link = "http://www.google.com";  
URL url = new URL(link);  
URLConnection conn = (URLConnection) url.openConnection();  
conn.connect();
```

Using:

```
InputStream is = conn.getInputStream();  
BufferedReader reader = new BufferedReader(  
    new InputStreamReader(is, "UTF-8"));  
String webPage = "", data="";  
while ((data = reader.readLine()) != null){  
    webPage += data + "\n";  
}
```



Example (1)

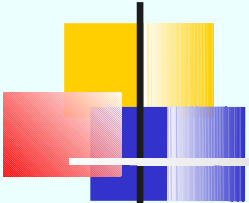
MainActivity.java:

```
public class MainActivity extends Activity {
    private EditText urlField;
    private TextView data;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        urlField = (EditText)findViewById(R.id.editText1);
        data = (TextView)findViewById(R.id.textView2);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }

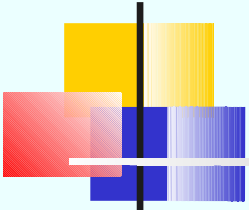
    public void download(View view){
        String url = urlField.getText().toString();
        new DownloadWebPage(this,data).execute(url);
    }
}
```



Example (2)

DownloadWebPage.java (1):

```
public class DownloadWebPage extends AsyncTask{
    private TextView dataField;
    private Context context;
    public DownloadWebPage(Context context,TextView dataField) {
        this.context = context;
        this.dataField = dataField;
    }
    //check Internet connection.
    private void checkInternetConenction(){
        ConnectivityManager check = (ConnectivityManager) this.context.
            getSystemService(Context.CONNECTIVITY_SERVICE);
        if (check != null){
            NetworkInfo[] info = check.getAllNetworkInfo();
            if (info != null)
                for (int i = 0; i <info.length; i++)
                    if (info[i].getState() == NetworkInfo.State.CONNECTED)
                        Toast.makeText(context, "Internet is connected",
                            Toast.LENGTH_SHORT).show();
        }
        else{
            Toast.makeText(context, "not conencted to internet",
                Toast.LENGTH_SHORT).show();
        }
    }
    protected void onPreExecute(){
        checkInternetConenction();
    }
}
```

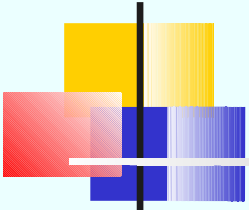


Example (3)

DownloadWebPage.java (2):

```
@Override
protected String doInBackground(String... arg0) {
    try{
        String link = (String)arg0[0];
        URL url = new URL(link);
        HttpURLConnection conn = (HttpURLConnection)url.openConnection();
        conn.setReadTimeout(10000);
        conn.setConnectTimeout(15000);
        conn.setRequestMethod("GET");
        conn.setDoInput(true);
        conn.connect();
        InputStream is = conn.getInputStream();
        BufferedReader reader = new BufferedReader(new InputStreamReader(
            (is, "UTF-8") ));
        String data = null;
        String webPage = "";
        while ((data = reader.readLine()) != null){
            webPage += data + "\n";
        }
        return webPage;
    }catch(Exception e){
        return new String("Exception: " + e.getMessage());
    }
}

@Override
protected void onPostExecute(String result){
    this.dataField.setText(result);
}
}
```



Example (4)

activity_main.xml:

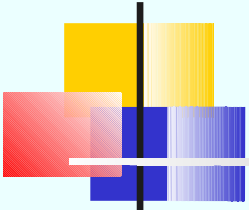
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" >

    <TextView android:id="@+id/textView1"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="30dp" android:text="@string/url"
        android:textAppearance="?android:attr/textAppearanceMedium" />

    <EditText android:id="@+id/editText1"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_below="@+id/textView1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="18dp" android:ems="10" />

    <Button android:id="@+id/button1"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_below="@+id/editText1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="69dp"
        android:onClick="download" android:text="@string/click" />

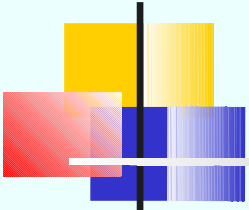
    <TextView android:id="@+id/textView2"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/editText1"
        android:layout_below="@+id/button1"
        android:layout_marginTop="56dp"
        android:text="@string/google"
        android:textAppearance="?android:attr/textAppearanceSmall" />
</RelativeLayout>
```



Example (5)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">NetworkConnection</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="url">URL here</string>
<string name="click">Download WebPage</string>
<string name="google">http://www.tutorialspoint.com</string>
</resources>
```



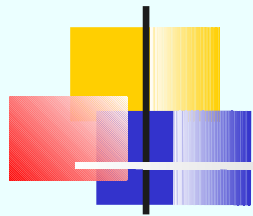
Example (6)

manifest.xml:

```
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission
  android:name="android.permission.ACCESS_NETWORK_STATE" />

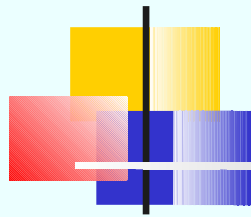
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.networkconnection.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```



Example (7)

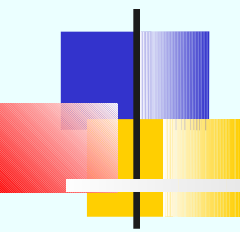




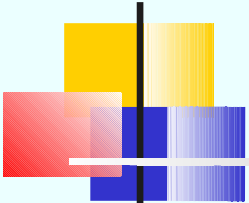
Exercises

Create an app:

- Enable to fetch keyword to google.com
- Enable to get content from that site and extract data to display



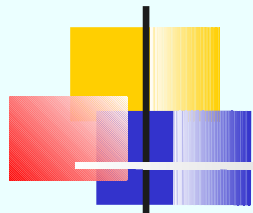
RSS parser



RSS document

Example:

```
<rss version="2.0">
<channel>
  <title>Sample RSS</title>
  <link>http://www.google.com</link>
  <description>World's best search engine</description>
</channel>
</rss>
```



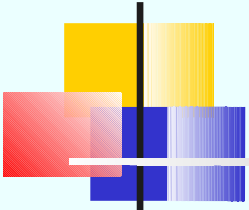
RSS parser

Create:

```
private XmlPullParserFactory xmlFactoryObject =  
XmlPullParserFactory.newInstance();  
private XmlPullParser myparser = xmlFactoryObject.newPullParser();
```

Using:

```
myparser.setInput(stream, null);  
int event = myParser.getEventType();  
while (event != XmlPullParser.END_DOCUMENT) {  
    String name=myParser.getName();  
    switch (event){  
        case XmlPullParser.START_TAG:  
            break;  
        case XmlPullParser.END_TAG:  
            if(name.equals("temperature")) {  
                temperature = myParser.getAttributeValue(null,"value")  
            }  
            break;  
    }  
    event = myParser.next();  
}
```



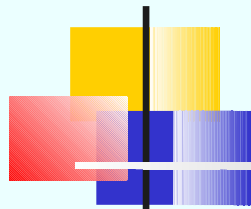
Example (1)

MainActivity.java:

```
public class MainActivity extends Activity {
    private String
    finalUrl="http://tutorialspoint.com/android/sampleXML.xml";
    private HandleXML obj;
    private EditText title,link,description;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        title = (EditText)findViewById(R.id.editText1);
        link = (EditText)findViewById(R.id.editText2);
        description = (EditText)findViewById(R.id.editText3);
    }

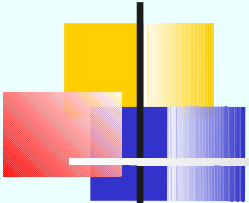
    public void fetch(View view){
        obj = new HandleXML(finalUrl);
        obj.fetchXML();
        while(obj.parsingComplete);
        title.setText(obj.getTitle());
        link.setText(obj.getLink());
        description.setText(obj.getDescription());
    }
}
```



Example (2)

HandleXML.java (1):

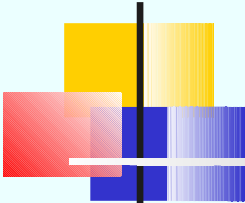
```
public class HandleXML {  
    private String title = "title";  
    private String link = "link";  
    private String description = "description";  
    private String urlString = null;  
    private XmlPullParserFactory xmlFactoryObject;  
    public volatile boolean parsingComplete = true;  
  
    public HandleXML(String url){  
        this.urlString = url;  
    }  
    public String getTitle(){  
        return title;  
    }  
    public String getLink(){  
        return link;  
    }  
    public String getDescription(){  
        return description;  
    }  
}
```



Example (3)

HandleXML.java (2):

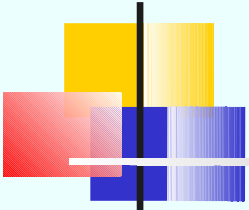
```
public void parseXMLAndStoreIt(XmlPullParser myParser) {
    int event; String text=null;
    try {
        event = myParser.getEventType();
        while (event != XmlPullParser.END_DOCUMENT) {
            String name=myParser.getName();
            switch (event){
                case XmlPullParser.START_TAG:
                    break;
                case XmlPullParser.TEXT:
                    text = myParser.getText();
                    break;
                case XmlPullParser.END_TAG:
                    if(name.equals("title")){
                        title = text;
                    }else if(name.equals("link")){
                        link = text;
                    }else if(name.equals("description")){
                        description = text;
                    }else{
                    }
                    break;
            }
            event = myParser.next();
        }
        parsingComplete = false;
    } catch (Exception e) {
        e.printStackTrace();
    }
}
```



Example (4)

HandleXML.java (3):

```
public void fetchXML(){
    Thread thread = new Thread(new Runnable(){
        @Override
        public void run() {
            try {
                URL url = new URL(urlString);
                HttpURLConnection conn = (HttpURLConnection)
                    url.openConnection();
                conn.setReadTimeout(10000 /* milliseconds */);
                conn.setConnectTimeout(15000 /* milliseconds */);
                conn.setRequestMethod("GET");
                conn.setDoInput(true);
                // Starts the query
                conn.connect();
                InputStream stream = conn.getInputStream();
                xmlFactoryObject = XmlPullParserFactory.newInstance();
                XmlPullParser myparser = xmlFactoryObject.newPullParser();
                myparser.setFeature(XmlPullParser.FEATURE_PROCESS_NAMESPACES,
                    false);
                myparser.setInput(stream, null);
                parseXMLAndStoreIt(myparser);
                stream.close();
            } catch (Exception e) {
            }
        }
    });
    thread.start();
}
```



Example (5)

activity_main.xml (1):

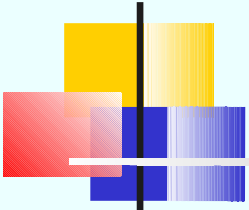
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent" android:layout_height="match_parent"
    tools:context=".MainActivity" >
```

```
<TextView android:id="@+id/textView1"
    android:layout_width="wrap_content" android:layout_height="wrap_content"
    android:layout_alignParentTop="true"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="26dp"
    android:text="@string/hello_world"
    android:textAppearance="?android:attr/textAppearanceLarge" />
```

```
<TextView android:id="@+id/textView2"
    android:layout_width="wrap_content" android:layout_height="wrap_content"
    android:layout_below="@+id/textView1"
    android:layout_marginTop="48dp"
    android:layout_toLeftOf="@+id/textView1"
    android:text="@string/title" />
```

```
<TextView android:id="@+id/textView3"
    android:layout_width="wrap_content" android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/textView2"
    android:layout_below="@+id/textView2"
    android:layout_marginTop="27dp"
    android:text="@string/link" />
```

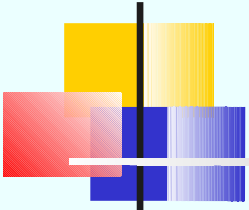
```
<EditText android:id="@+id/editText1"
    android:layout_width="wrap_content" android:layout_height="wrap_content"
    android:layout_alignBaseline="@+id/textView2"
    android:layout_alignBottom="@+id/textView2"
    android:layout_alignParentRight="true"
    android:ems="10" >
<requestFocus />
</EditText>
```



Example (6)

activity_main.xml (2):

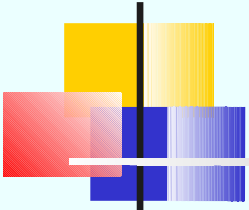
```
<EditText
android:id="@+id/editText2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/textView3"
android:layout_alignBottom="@+id/textView3"
android:layout_alignLeft="@+id/editText1"
android:ems="10" >
</EditText>
<EditText
android:id="@+id/editText3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/textView4"
android:layout_alignBottom="@+id/textView4"
android:layout_alignLeft="@+id/editText2"
android:ems="10" />
<Button
android:id="@+id/button1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@+id/editText3"
android:layout_centerHorizontal="true"
android:layout_marginTop="37dp"
android:onClick="fetch"
android:text="@string/fetch" />
<TextView
android:id="@+id/textView4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentLeft="true"
android:layout_centerVertical="true"
android:text="@string/description" />
</RelativeLayout>
```



Example (7)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">RSSReader</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Sample RSS Reader</string>
<string name="title">title</string>
<string name="link">link</string>
<string name="description">Description</string>
<string na
```



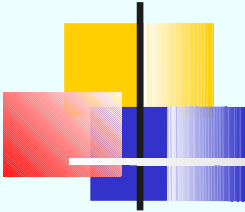
Example (8)

manifest.xml:

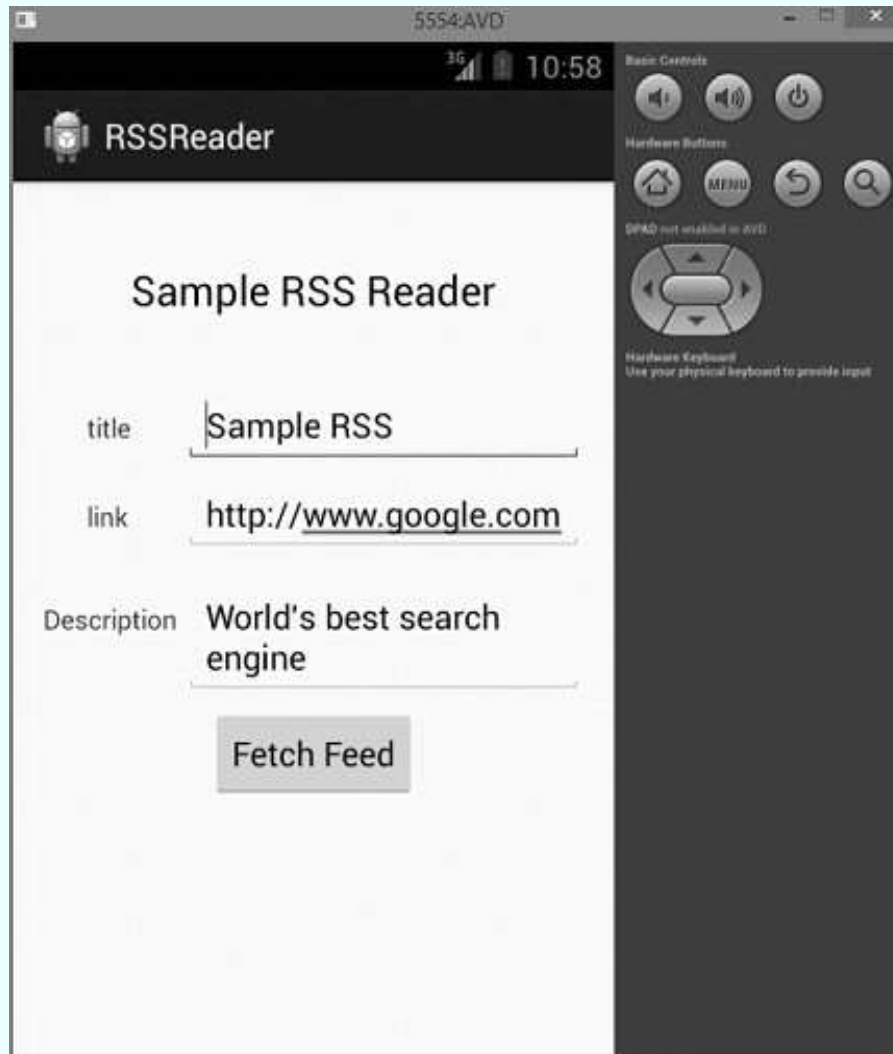
```
<uses-permission android:name="android.permission.INTERNET"/>

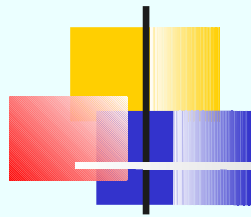
<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >

    <activity
        android:name="com.example.rssreader.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```



Example (9)

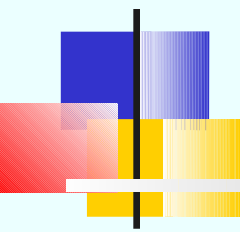




Exercises

Create an app:

- Enable to fetch RSS from a news site
- Enable to get content from that site and extract data to display



Questions?
